

Logistic Regression

Tuesday, 1 April 2025 2:19 PM

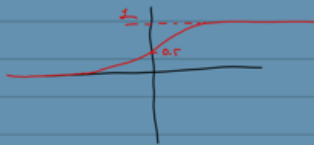


Logistic regression...

$$\Rightarrow f(z) = \frac{1}{1+e^{-z}}, f(wx) = \frac{1}{1+e^{-wx}}$$

$$\text{as } z \rightarrow \infty \Rightarrow f(z) = 1$$

$$z \rightarrow -\infty \Rightarrow f(z) = 0.$$



* Logistic Regression :- (Another form of Naive Bayes on gaussian distribution).

$$N.B :- P(x|y) \cdot P(y).$$

$$L.R :- P(y|x), \text{ where the assumption of } f_{\text{ids}}$$

$$\text{that is follows } P(y|x) = \frac{1}{1+e^{-wx+b}}$$

NLE

$$w, b = \text{argmax}_{w, b} \prod_{i=1}^n P_x(y_i | \vec{x}_i = w, b).$$

$$= \text{argmax}_{w, b} \sum_{i=1}^n \log P_{w, b}(y_i | \vec{x}_i, w)$$

$$= \text{argmax}_{w, b} \sum_{i=1}^n \log \left(\frac{1}{1+e^{-y_i w x_i}} \right).$$

$$= \text{argmax}_{w, b} - \sum_{i=1}^n \log (1+e^{-y_i w x_i}).$$

$$= \text{argmin}_{w, b} \sum_{i=1}^n \log (1+e^{-y_i w x_i}).$$

3. Conclusion

By setting $\tilde{y} = 2y - 1$, the binary-cross-entropy loss for $y \in \{0, 1\}$

$$-[y \log \sigma(z) + (1-y) \log(1-\sigma(z))]$$

is identical to the logistic-loss

$$\log(1+e^{-\tilde{y}z}) \text{ for } \tilde{y} \in \{-1, +1\}.$$

MAP :-

$$\text{argmax } P(w|D) \propto \underbrace{P(D|w)}_{MLE} \cdot \underbrace{P(w)}_{\text{Prior.}}$$

↳ usually taken as a normal distribution with 0 mean & some standard deviation.

$$P(w) = \mathcal{N}(0, \sigma^2).$$

⇒ You'll get exactly the same formula, but you'll get an extra regularization term.

$$= \arg \min_w \sum_{i=1}^n \log(1 + e^{-y_i w^T x_i}) + \lambda w^T w.$$

⇒ Watch demo in Lec 11: time 45:00.

⇒ we had

$$P(x|y) \cdot P(y) = P(y|x) = \frac{1}{1 + e^{-w^T x y}}$$

$$MLE = \max_w \prod P(y_i | x_i) \dots$$

$$= \min_w \sum_{i=1}^n \log(1 + e^{-w^T x_i y_i}).$$

Convex fⁿ.

* Implementation:-

$$f(z) = \frac{1}{1 + e^{-z}}.$$

$$f'(z) = \frac{-1}{(1 + e^{-z})^2} (-e^{-z}) \quad \} N \times 1$$

$$= \frac{e^{-z}}{(1 + e^{-z})^2}$$

$$= \frac{1 + e^{-z} - 1}{(1 + e^{-z})^2}$$

$$= \frac{f(z) - f(z)^2}{f(z)(1 - f(z))} \quad \} N \times 1$$

$N \times 1 \quad N \times 1$

$$L(\hat{y}, y) = -y \log a - (1 - y) \log (1 - a)$$

$\hat{y} \Rightarrow$ predicted

$$21. \quad (-1 + 1 - 1) \quad \checkmark$$

$$\frac{\partial L}{\partial a} = \left(\frac{-y}{a} + \frac{(1-y)}{(1-a)} \right)$$

$$\frac{\partial L}{\partial w} = \frac{\partial L}{\partial a} \cdot \frac{\partial a}{\partial z} \cdot \frac{\partial z}{\partial w}$$

$$z = w^T x + b \quad \frac{\partial z}{\partial w} = x \quad \text{--- (1)}$$

$$\frac{\partial z}{\partial b} = 1$$

$$\frac{\partial a}{\partial z} = a(1-a).$$

$$\frac{\partial L}{\partial a} = \frac{-y}{a} + \frac{(1-y)}{(1-a)}$$

$$\frac{\partial L}{\partial w} = \frac{\partial L}{\partial a} \cdot \frac{\partial a}{\partial z} \cdot \frac{\partial z}{\partial w}$$

$$\therefore \frac{\partial L}{\partial w} = \left(\frac{-y}{a} + \frac{(1-y)}{(1-a)} \right) (a(1-a)) (x).$$

$$= (-y(1-a) + (1-y)a) (x)$$

$$= (-y + ay + a - ay) (x)$$

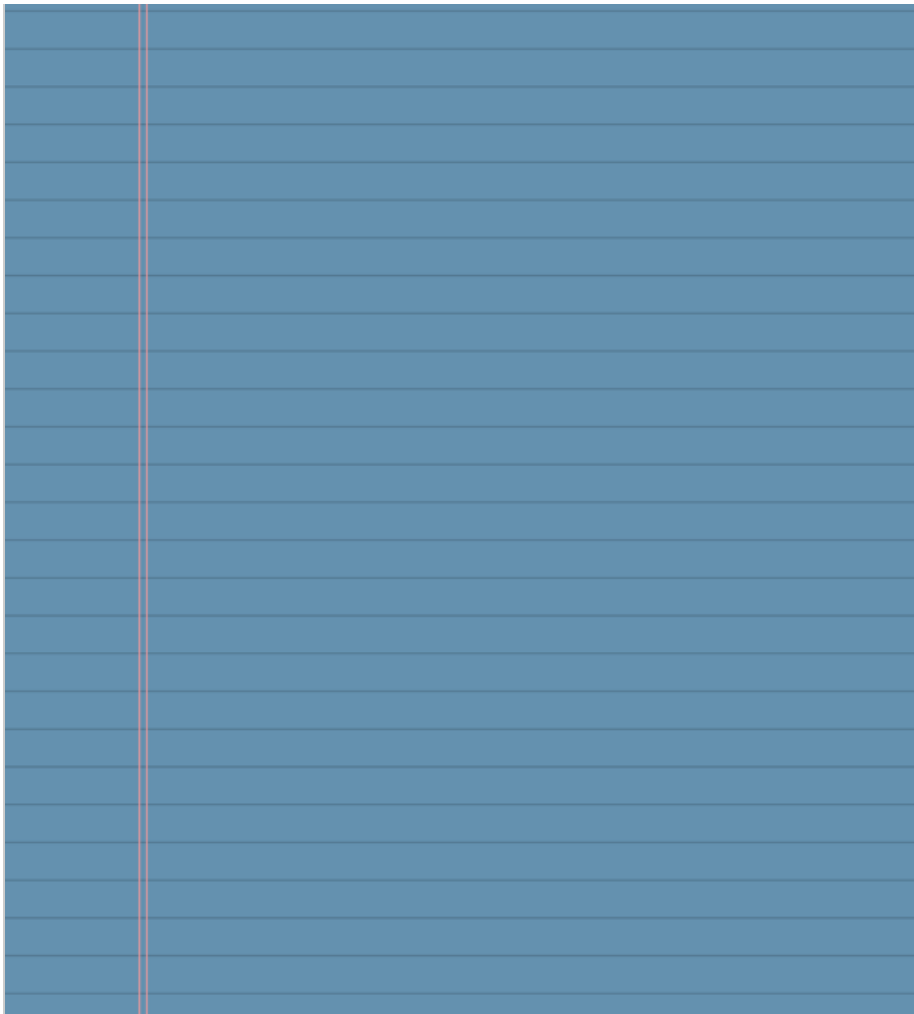
$$\boxed{\begin{aligned} \frac{\partial L}{\partial w} &= (a-y) \quad \begin{matrix} N \times 1 & m \times N \end{matrix} \\ \frac{\partial L}{\partial b} &= (a-y) \end{aligned}}$$

$$x \Rightarrow m \times n$$

$$w \Rightarrow n \times 1$$

$$b \Rightarrow n \times 1$$

$$w^T x + b$$



Why is it called a regression?

<https://stackoverflow.com/questions/30499018/why-is-logistic-regression-called-regression>

Why is the decision boundary linear?

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The probability is greater than 0.5 when:

$$g(w) > 0.5$$

and this is true when:

$$w = K^T X \geq 0$$

The hyperplane:

$$K^T X = 0$$

is the decision boundary which is linear.